CS 440, Introduction to Software Engineering, Course Policies
Fall 2026

Course Objectives: At the end of this course, students will be able to:

- Participate fully as a valuable team member on large complex software projects.
- Explain the overall management concepts and issues of large software projects, including the full software life cycle of large projects, from project inception through retirement.
- Specify the requirements for a software product.
- Design a software system, using a variety of software design strategies.
- Implement a software design developed by others, using agile methods & modern CASE tools.
- Verify and validate software systems.

Primary ABET Learning Outcomes:

- An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
- An ability to communicate effectively with a range of audiences.
- An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Course Description: Software life-cycle model, requirement specification techniques, large-scale software design techniques and tools, implementation issues, testing and debugging techniques, software maintenance. 3 credits for undergraduates, 4 for graduates. Technical elective, required for the SE concentration.

Instructor: John Bell, jbell@uic.edu,
921 SEO, 413-9054
http://www.cs.uic.edu / ~jbell
Office Hours: MW 3:00-4:00 ARC 240 or by appointment.

Teaching Assistants: Jade Qiu, jqiu23@uic.edu
Other grad and undergrad TAs & Scrum Master – TBD
Office Hours: By request - See web for details.

Meeting Times: MWF in ARC 242 at 1:00-1:50 and in ARC 240 at 2:00-2:50 (2 sections)
Mandatory in-person on-campus attendance.

Course Web Page: http://www.cs.uic.edu/~i440

Prerequisites: CS 342

If you are an undergraduate student who does not have the necessary pre-requisites, DROP THE CLASS NOW. Otherwise you will be automatically dropped later, when it will be too late to sign up for anything else instead. **Graduate students are not allowed to take CS 440 after or concurrently with CS 442.** This course requires a significant amount of computer programming, as part of a team project, most likely in C++ or Java. All students are expected to know at least one of these languages, regardless of background or graduate status.

Required Textbook: Bernd Bruegge and Allen H. Dutoit, "Object-Oriented Software Engineering - Using UML, Patterns, and Java", Third Edition, Prentice Hall, ISBN-10: 0136061257, ISBN-13: 9780136061250

Required Resources:
Blackboard: https://uic.blackboard.com
Piazza: http://www.piazza.com
Gradescope: http://www.gradescope.com
O'Reilly Technical Books: http://learning.OReilly.com

Other Good Books:

1. **Fowler, Martin and Kendall Scott, "UML Distilled, 3rd Edition".**
2. **Gamma Erich, et. al., "Design Patterns – Elements of Reusable Object-Oriented Software" (Available on UIC's Safari system.)**
3. **Robertson & Robertson, "Mastering the Requirements Process" (Also on UIC Library's Safari system.)**
4. Pezze & Young, "Software Testing and Analysis"
5. Strunk & White, "The Elements of Style, 4th Ed."

Required In-Person Attendance

- **Attendance during regular class lectures is expected, and will be checked.** Any student missing a regular class period is required to watch the class recording within 48 hours, and is responsible for all material presented in class.
- Once groups are formed, then each group should have a representative present at each regular class period. Those who cannot attend should inform their team representative as to the reason for their absence.
- Individuals or groups with excessive absences will be asked to meet with the instructor to discuss the situation.
- Groups are required to meet **physically face-to-face** at least once a week starting in the third week. Experience has shown (many times) that groups perform MUCH better when they have more in-person interactions, as opposed to online meetings or even worse, e-mail/discord communications. **Zoom, Discord, Slack, etc. does not count.**
- There are four weeks during the term (6th, and 13th to 15th) during which student presentations will be made and during which 100% attendance is required.
- In-class group exercises may occur at any time during the term, to be counted towards class participation.

Classroom Conduct Policy

Students in this class are expected to behave as responsible, mature, courteous adults. This includes:

- Arriving on time and not departing early, to the extent possible.
- Paying attention in class.
- Actively and productively engaging in classroom discussions and activities during group activity time.
- Not engaging in side conversations or other activities not related to the current class activities, and not distracting fellow students or the instructor.
- Refraining from comments or other activities that could be considered offensive, demeaning, inconsiderate, racist, or otherwise inappropriate.

Graduate Student 4th Credit

In order to receive the 4th credit, graduate students must complete and present a short research paper on a topic to be agreed upon related to software engineering.

Planned Schedule

A detailed schedule of activities and assignment deadlines will appear on the course web site, to be updated as the course progresses. In addition, the following exam dates are expected, subject to conflicts:

NO CLASS: Monday September 7th – Labor Day and November 25th-27th - Thanksgiving.
MIDTERM EXAM: In week 9 covering weeks 1-8.
Combined section evening exam, exact date depending on room availability.
FINAL EXAM: Combined section exam during finals week. Exact date and time TBD.

Term Project(s)

This course is primarily concerned with large complex software engineering projects which require multiple developers and significant time to complete. In order to practice these concepts, a large group project will be conducted throughout the term. This project will serve as the focal point for all material covered in class. Periodic assignments will be due throughout the term, including intermediary results and progress reports.

This semester you will actually complete one-half each of two projects, in parallel: You will **plan and design** a project to be implemented at some time in the future, and you will **implement a working prototype** based on a project previously planned and designed by former students.

Individual Accountability for Group Work

As a general rule, everyone is expected to do their best on group project work, and to contribute their fair share. Initial point distributions will be based on the assumption that all members of a group are contributing equally, to the best of their abilities.

Unfortunately, however, that has not always been the case, and so therefore some measures will be taken to measure and monitor individual accountability for group work, as follows:

- Group work submissions should only include the names of those members who actually contributed to their production.
- Weekly progress/activity reports should indicate ***for each individual*** what they accomplished since the previous week, and what they will be working on during the upcoming week. These need to document specific accomplishments and activities, not generalities, and should show progress towards a goal over the course of multiple weeks.
- All members of a group should be submitting their individual work to the GitHub repository as it is accomplished, e.g. for the coding project.
- All members of a group should be active in a specified SW project management tool, taking on tasks that need to be done, and marking them complete when they have been completed.
- Periodically all students will be asked to complete peer evaluations, appraising how each of their teammates has contributed, as well as appraising their own efforts. Note that this is based on effort and good teamsmanship, not accomplishments, as some students have greater abilities than others.
- **Low contribution may result in significant reduction in group scores for non-contributing individuals, leading to a grade reduction of one or more letter grades, possibly failing the course.**
- Periodic group conferences with the instructor may be arranged, on a semi-random basis, depending on needs.

Grading Policy

Numerical scores will be based upon the following contributions:

Individual Work

Midterm Exam (90 minutes)	15 points	≥50 points
Final Exam (2 hours)	20 points	
Initial project brainstorming report	5 points	
Presentation Evaluations	5 points total	
Weekly Progress/Activity reports	5 points total	
Individual work TBD and class participation	as needed	

Design Project

Design Project Description and Requirements Reports	10 points each	50 points
Final Design Project Report	20 points	
(2) Oral Presentations - Description and Final	5 points each	

Implementation Project

(2) Scenarios	5 points each	50 points
(2) Demonstrations to TA	10 points each	
Final Oral Presentation & Demo	5 points	
Final Implementation Project Report	10 points	
CASE tool usage (Jira)	5 points total	

Other assignments and/or quizzes	as needed
Total:	Normalized to 100 point scale

- At any time during the semester your Percentage of Possible Points, PPP, can be calculated as:

$$PPP = \frac{Points\ accumulated\ to\ date}{Possible\ points\ to\ date} * 100\%$$

- Mapping of PPP score to letter grades will not necessarily follow the traditional 90/80/70/60 grade breaks, however some effort will be made to keep it close. Detailed feedback showing each student's place relative to the rest of the class will be made after each of the midterm exams is graded and returned. Separate grading curves may be applied to each TA.
- **Regardless of overall scores, passing a majority of exams and completing a majority of assignments with good-faith effort are required to be eligible for a passing grade.**

Grading Guidelines

1. Reports will initially be given a base score of 90 for excellent work, 80 for very good work, 70 for good work, 60 for fair work, 50 or lower for poor work.
2. Then up to 9 individual points may be either added to or subtracted from the base score to refine scores to a finer resolution. In general, higher quality work earns higher scores.
3. The value of a report can also be considered according to the following expression:

$$Value(\ Report\ ) \propto \frac{Quantity(\ Information\ ) * Quality(\ Information\ )}{Effort\ required\ to\ comprehend\ the\ information}$$

Exam Policy

- Any exam conflict needs to be brought to the instructor's attention for resolution **before** the regularly scheduled exam. Requests for make-up exams after the regularly scheduled exam will not normally be granted.
- Exams will be closed-book and closed-notes.
- All exams will be cumulative, with emphasis on material which has not been covered on previous exams.
- All material covered in class or in assigned reading or which should have been learned in the course of completing homework or contributing to projects is fair game on exams. No more specific information will be provided as to exam content.
- Anyone who fails to stop working on their exam when time is called will receive a minimum of a five point late penalty.

Academic Integrity

Students are encouraged to study together and to help each other learn. When one student teaches another, both benefit from the experience.

However, **it is a strict violation of class and university policy for any student to hand in any work that is not 100% their own creation**. Therefore:

- All work submitted must be 100% created by the author(s) listed on the submission.
  - Group work submissions should only include the names of those members who actually contributed to their production.
  - Individual assignments are to be conducted individually, not in collaboration with others.
  - **Copying of any material from the Internet or elsewhere is expressly forbidden.**
  - **The use of Chegg, ChatGPT, or other AI tools or sources is expressly forbidden.** (Spelling and grammar checkers are acceptable. See also the AI policy given below.)

- Falsification of meeting/activity reports, attendance records, or other similar documentation are ethical violations. (**DO NOT** sign the attendance sheet for anyone not present.)
- Students may discuss homework problems, including background concepts and general solution strategies, but they are forbidden from discussing or sharing specific solutions. In particular, it is forbidden for any student to show any other student any portion of their computer programs or homework solutions for any reason, including debugging assistance. This means you must hand in your own homework. You are not allowed to see anyone else's work, or show your work to anyone. Failure to protect the privacy of your work may be a violation.
- All transgressions will be reported to the Office of Student Judicial Affairs, http://www.uic.edu/depts/sja, for the permanent record and possibly for further action. Consequences may range from a zero on the assignment to expulsion from the U of I System, depending on the severity of the transgressions.
  - All parties involved will be reported, both the original source and those who copied.
  - Failure to protect the privacy of your work may be a violation.
- In the case of extreme discrepancy between homework performance and exam performance (e.g. very high homework scores and very low exam scores), the instructor shall determine which scores more accurately reflect the students' true work.

AI Policy

(Generative) AI is becoming more and more prevalent, powerful, and useful in many domains. However: (1) Used entirely on its own, it can produce some truly awful results, and (2) Noone really learns much by having a tool do all their work for them. Therefore, the following policies are in place for CS 440:

- The first draft of any written material or code must be human generated. The original creativity is the responsibility of the human.
- The final submitted material must be carefully reviewed and edited by a human. **DO NOT** hand in any AI-generated material without a human reviewing it carefully first.
- In between the first and final drafts, AI may be used to proofread, refine, polish, etc. the material first created by a human.
- All code (and other written material) must be fully understood by everyone on the team. If AI produces code or other material that you don't understand, don't use it.

Special Considerations

- **Requests for Consideration** should be submitted via the online form available on Blackboard, e.g. for academic conflicts, hospital stays, or other special circumstances.
- Assignments submitted after the due date and time, but within 24 hours, will be assessed a 20% penalty. No assignments will be accepted more than 24 hours after the due date / time.
- All appeals for grading errors, no matter how justified, must be submitted within two weeks after graded assignments are returned. No appeals for regrades will be heard after that time.
- **Any student that does not participate fully in their project activities may have their project scores reduced at the end of the term, which will result in a grade reduction.**

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

Following campus policy, if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please use the Request for Consideration form available on Blackboard to request accommodations.

Valuable Campus Resources

Academic Success

- UIC Tutoring Resources
- College of Engineering tutoring program
- Equity and Inclusion in Engineering Program
- UIC Library and UIC Library Research Guides.
- Offices supporting the UIC Undergraduate Experience and Academic Programs.
- Student Guide for Information Technology

Wellness

- **Counseling Services**: You may seek free and confidential services from the Counseling Center at https://counseling.uic.edu/.
- Access U&I Care Program for assistance with personal hardships.
- **Campus Advocacy Network**: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at http://can.uic.edu/.

Safety

- UIC Safe App—PLEASE DOWNLOAD FOR YOUR SAFETY!
- UIC Safety Tips and Resources
- Night Ride
- Emergency Communications: By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.